



COURSE SLIDES

AI for Retail

Course Code: C398

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v1.0 · 17 August 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

ACLP-certified trainer; delivers applied AI and data courses for business teams.

DELIVERS

Practical, no-code AI courses for retail, marketing, operations and analytics roles.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

1

Your name, your store or organisation, and your role.

2

What you sell, and how many outlets or channels you run.

3

Which AI tools you have already tried at work, if any.

4

The one retail task you would most like to hand to AI by this evening.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

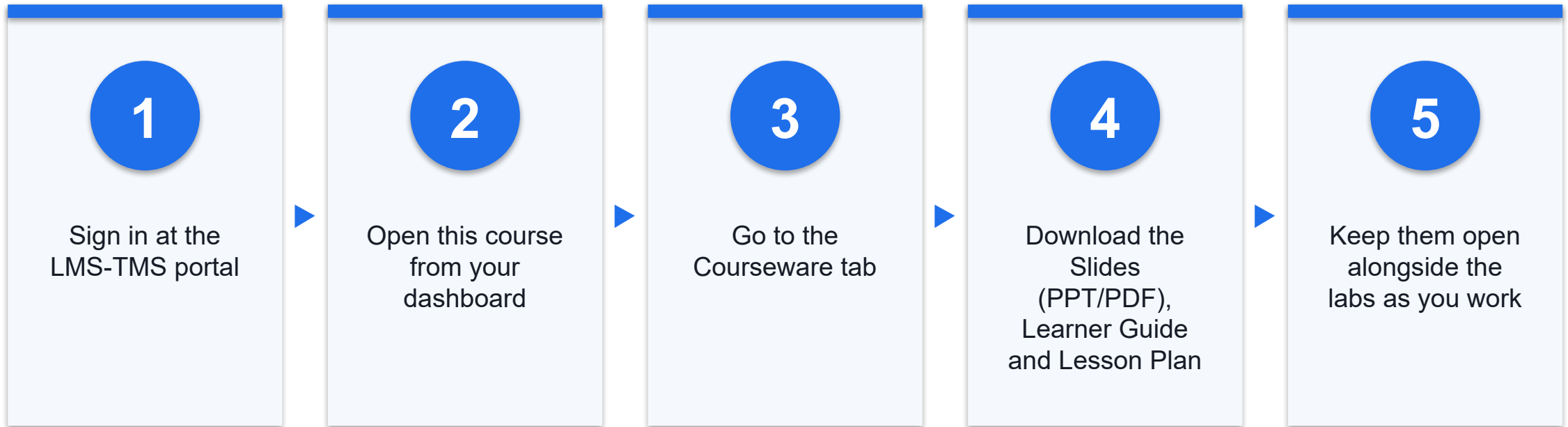
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Download Course Material



Portal: <https://lms-tms.tertiaryinfotech.com>

Lesson Plan — 1 Day, 7.5 instructional hours/day

Day 1 - Topics Covered

- AI foundations for retail, then AI applied across operations and customer experience
 - Topic 1: Introduction to AI for Retail (Labs 1–4)
 - Topic 2: Applying AI to Retail Operations and Customer Experience (Labs 5–9)
- Hands-on lab time
 - 9 labs · 330 minutes · one connected retail scenario

How the Day Runs

- Daily timing
 - 9:30am-6:30pm · 1-hour lunch · tea breaks within training time
- How the day runs
 - Concepts and trainer demo, then you build it yourself in the lab.
 - Every lab ends with a 'Test it' check you run yourself.
 - Recap and Q&A against the learning outcomes to close.

Learning Outcomes

1

AI Landscape

Describe the AI landscape from generative AI to AI agents, and identify high-value AI use cases across the retail value chain.

2

AI Assistants

Use AI assistants such as ChatGPT, Claude, Microsoft Copilot and Google Gemini to complete everyday retail tasks with well-structured prompts.

3

Content & Visuals

Generate on-brand product descriptions, marketing copy and product visuals with AI, and check them before they reach a customer.

4

Responsible AI

Apply customer data privacy, governance and responsible-AI practices when using AI tools on retail data.

5

Service & Personalisation

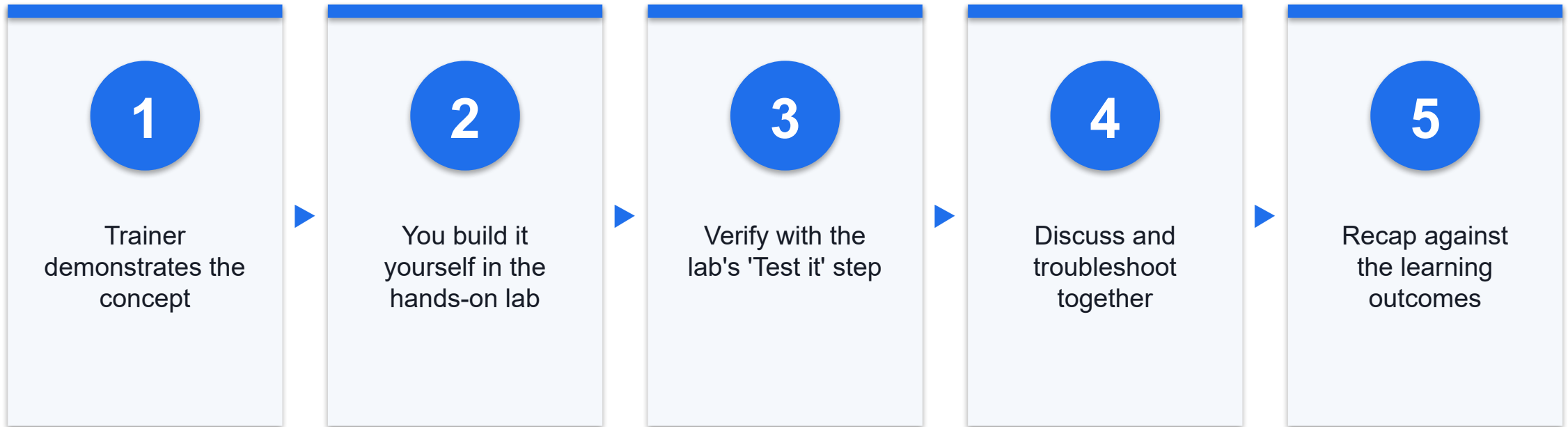
Deploy AI for customer service, personalised recommendations and targeted promotions in a retail setting.

6

Analytics & Workflows

Apply AI to demand forecasting, inventory planning, pricing, merchandising and sales analytics, and build simple AI workflows for daily retail tasks.

How You'll Learn





CORE CONCEPTS

AI Foundations for Retail

Key Concepts

1

Generative AI

Creates new text, images and copy from a prompt. The engine behind product descriptions and campaign content.

2

AI Assistant

A chat interface - ChatGPT, Claude, Copilot, Gemini - that you brief in plain English to do a retail task.

3

AI Agent

Goes beyond answering: plans and carries out a multi-step task using tools and your data.

4

Grounding

Giving the AI your own FAQ, policies and product data so answers come from your store, not from the open internet.

5

Prompt Pattern

Role, context, task, format, constraints - the five-part brief that makes AI output usable first time.

6

Human in the Loop

A person checks and approves every AI output that reaches a customer, a price or a purchase order.

The Retail AI Value Chain

1

Source & Buy

Supplier research, spec comparison and negotiation prep.

2

Merchandise & Price

Assortment reviews, markdown candidates, category narratives.

3

Market & Sell

Product copy, campaign assets, social captions, email.

4

Serve

FAQ assistants, returns handling, review replies, tone control.

5

Plan & Forecast

Demand forecasts, reorder points, slow-mover and stock-risk reports.

6

Analyse

Plain-English sales analytics, KPI summaries and action lists.

WHY THIS COURSE

**AI does not replace the retailer's judgement
- it removes the hours before the
judgement.**

Every lab today produces something you could use in your store tomorrow: product copy, a service assistant, a forecast, a pricing review, a working workflow.

What You'll Build

Content & Brand

- A reusable five-part house prompt template
- AI product descriptions and SEO metadata for a catalogue
- Campaign copy and an AI-generated product visual

Customer Experience

- A store service assistant grounded in your own policies
- Four customer segments with a targeted offer each
- Next-best-product recommendations for named customers

Operations & Analytics

- A three-month demand forecast and reorder plan
- A margin, markdown and category review
- A no-code daily sales-summary workflow

How Every Lab Progresses

1

Start from a real retail artifact - a product sheet, a sales export, a customer enquiry.

2

Brief the AI with the five-part prompt pattern instead of a one-line question.

3

Generate a first draft, then challenge it: check the numbers, the policy and the tone.

4

Verify with the lab's 'Test it' step so you know the output is fit to use.

5

Save the working prompt into your own prompt library so the next run takes minutes.

TOPIC 01

Introduction to AI for Retail

The AI landscape · use cases across the retail value chain · hands-on with ChatGPT, Claude and Copilot · AI-generated product and marketing content · customer data privacy and responsible AI

Key Concepts — Introduction to AI for Retail

1

Generative AI creates new content - text, images, copy - from a prompt. An AI agent goes further: it plans and carries out a multi-step task using tools and data.

2

Every stage of the retail value chain has an AI use case: sourcing, merchandising, pricing, marketing, store operations, e-commerce, service and after-sales.

3

An AI assistant is only as good as the brief. Role, context, task, format and constraints turn a vague answer into copy you can actually publish.

4

ChatGPT, Claude, Microsoft Copilot and Google Gemini answer the same brief differently. Comparing them on your own task is how you choose the right one.

5

AI cuts content production from hours to minutes for descriptions, campaigns and visuals - but every output needs a human check before a customer sees it.

6

Customer data is personal data. Anonymise before you paste, keep payment and identity data out of public AI tools, and keep a person accountable for the decision.

Hands-On Labs — Introduction to AI for Retail

Labs 1–2

- Brief an AI Assistant Like a Retail Manager
- Generate a Product Catalogue with AI

Labs 3–4

- Campaign Copy and AI Product Visuals
- Customer Data Privacy and Responsible AI

Brief an AI Assistant Like a Retail Manager

LAB 1

Three AI assistants, one retail brief. You will send a deliberately vague request, see what comes back, then rewrite it with the five-part pattern - role, context, task, format, constraints - and run the improved brief across ChatGPT, Claude and Copilot or Gemini. You finish with a house prompt template you will reuse in every later lab.

You'll build

A reusable five-part house prompt template, plus a scored comparison of three AI assistants on the same retail brief.

Tools: ChatGPT, Claude, Microsoft Copilot or Google Gemini, [labs/resources/products.csv](#)

Brief an AI Assistant Like a Retail Manager

STEP 1 OF 6



1

Open ChatGPT, Claude and either Microsoft Copilot or Google Gemini in three browser tabs and sign in to each.

Brief an AI Assistant Like a Retail Manager

STEP 2 OF 6

2

Send this deliberately vague one-line brief to ChatGPT, and read the answer critically.

PROMPT · paste into your AI assistant

Write a product description for a coffee mug.

Brief an AI Assistant Like a Retail Manager

STEP 3 OF 6

3

Rewrite the same request with the five-part pattern and send it again in a NEW chat.

PROMPT · paste into your AI assistant

Role: you are a senior retail copywriter for Bloom & Brew, a specialty home, gift and coffee retailer.

Context: the product is SKU BB-DRK-014, a 350ml double-walled stoneware mug, matte glaze, dishwasher safe, retail SGD 32.

Task: write the web product description.

Format: 60 words of body copy, then exactly five feature bullets of no more than 8 words each.

Constraints: British English, warm but not cute, no superlatives, no claims that are not in the context above.

Brief an AI Assistant Like a Retail Manager

STEP 4 OF 6



4

Send that identical five-part prompt to Claude and to Copilot or Gemini, without changing a single word.

Brief an AI Assistant Like a Retail Manager

STEP 5 OF 6



5

Score the three outputs 1-5 on brand fit, factual accuracy, format compliance and how long you would spend editing.

Brief an AI Assistant Like a Retail Manager

STEP 6 OF 6

6

Save your five-part prompt as a house template with placeholders, ready to reuse for any product.

PROMPT · paste into your AI assistant

Role: you are a senior retail copywriter for <YOUR STORE NAME>, a <YOUR CATEGORY> retailer.

Context: the product is <SKU>, <KEY SPECS>, retail <PRICE>.

Task: <THE ONE THING YOU WANT DONE>.

Format: <EXACT SHAPE OF THE ANSWER - length, sections, table columns>.

Constraints: <BRAND VOICE>, no claims outside the context above, say so if information is missing.

Brief an AI Assistant Like a Retail Manager

Test it

Open a fresh chat, paste your house template, and fill the placeholders with a DIFFERENT product from products.csv. The answer must come back in exactly the format you specified - the right length and the right number of bullets - with no correction needed from you. If you had to fix the shape of the answer, your Format and Constraints lines are still too loose: tighten them and run it again.

Generate a Product Catalogue with AI

LAB 2

Turn a row of a spreadsheet into web-ready product content. You will ground the assistant in the Bloom & Brew catalogue and brand voice guide, generate descriptions, feature bullets and SEO metadata for five SKUs in a single pass, then fact-check every claim back to the source file and fix the tone in one correction prompt rather than by hand.

You'll build

Web-ready descriptions, feature bullets, SEO titles and meta descriptions for five SKUs, in brand voice, as a table you can paste into a product feed.

Tools: ChatGPT or Claude, labs/resources/products.csv, labs/resources/brand_voice.md

Generate a Product Catalogue with AI

STEP 1 OF 6



Open products.csv and brand_voice.md, and choose five SKUs from at least three different categories.

Generate a Product Catalogue with AI

STEP 2 OF 6



2

Attach both files to a new chat so the assistant writes from your catalogue instead of from guesswork.

Generate a Product Catalogue with AI

STEP 3 OF 6

3

Ask for description, bullets and SEO metadata for all five SKUs in one table, in one pass.

PROMPT · paste into your AI assistant

Role: you are a senior retail copywriter for Bloom & Brew.

Context: use the attached products.csv for the facts and brand_voice.md for the tone. Nothing else.

Task: write web copy for these five SKUs: <SKU1>, <SKU2>, <SKU3>, <SKU4>, <SKU5>.

Format: one markdown table. Columns: SKU | Description (55-65 words) | Five feature bullets | SEO title (max 60 characters) | Meta description (max 155 characters).

Constraints: British English; use only facts present in products.csv; no invented materials, origins or awards; no superlatives; if a fact is missing write MISSING.

Generate a Product Catalogue with AI

STEP 4 OF 6



4

Fact-check every claim: read each description against the SKU's row and strike out anything the file does not support.

Generate a Product Catalogue with AI

STEP 5 OF 6

5

Correct the tone in one pass by naming what is wrong, instead of rewriting the copy yourself.

PROMPT · paste into your AI assistant

Rows 2 and 4 drift from the brand voice guide: they use superlatives and exclamation marks.

Rewrite **ONLY** those two descriptions to match brand_voice.md. Keep the word count, the bullets and the metadata unchanged.

Return the corrected rows only, in the same table format.

Generate a Product Catalogue with AI

STEP 6 OF 6



6

Paste the finished table into your product sheet, and save the prompt that produced it into your prompt library.

Generate a Product Catalogue with AI

Test it

Pick one finished description and trace every claim in it to a column in products.csv. If a claim has no column behind it, the copy is not publishable - delete or correct it, then add the missing rule to your Constraints line so the next batch does not repeat it. A clean pass means every sentence is supported by data.

Campaign Copy and AI Product Visuals

LAB 3

Build one seasonal promotion end to end. You will brief the campaign from the Bloom & Brew catalogue, generate email subject lines and body copy, three social captions and an in-store poster headline, then create a product visual with an image model and iterate it one variable at a time. The lab ends with the pre-publication check every AI-assisted campaign needs.

You'll build

A complete mini-campaign - email, three social captions, a poster headline and an AI-generated product visual - for one seasonal promotion.

Tools: ChatGPT or Claude, an image model (ChatGPT image generation, Gemini or Copilot Designer), [labs/resources/products.csv](#)

Campaign Copy and AI Product Visuals

STEP 1 OF 6



Choose the promotion: pick one category from products.csv, set the offer and the dates, and name the audience.

Campaign Copy and AI Product Visuals

STEP 2 OF 6

2

Brief the whole campaign in one prompt so every asset shares the same offer and voice.

PROMPT · paste into your AI assistant

Role: you are a retail marketing lead for Bloom & Brew.

Context: promotion is <OFFER> on <CATEGORY>, running <START DATE> to <END DATE>, aimed at <AUDIENCE>. Product facts come from the attached products.csv; tone from brand_voice.md.

Task: produce the full campaign kit.

Format: (1) three email subject lines, each max 45 characters; (2) one email body of 120-150 words with a single call to action; (3) three social captions, each under 220 characters with two hashtags; (4) one in-store poster headline of no more than six words.

Constraints: state the offer and the end date in every asset; no superlatives; no claims outside products.csv; British English.

Campaign Copy and AI Product Visuals

STEP 3 OF 6

3

Push the subject lines further by asking for distinct angles rather than more variants.

PROMPT · paste into your AI assistant

Rewrite the three subject lines so each takes a DIFFERENT angle: one urgency, one benefit, one curiosity. Keep each under 45 characters and include the character count in brackets after each line.

Campaign Copy and AI Product Visuals

STEP 4 OF 6

4

Generate the product visual with an image prompt that names the subject, setting, light and framing.

PROMPT · paste into your AI assistant

Create a photorealistic lifestyle image for a retail promotion.

Subject: a <PRODUCT> on a pale oak table, styled with a linen napkin and dried eucalyptus.

Setting: a bright Scandinavian-style cafe corner, soft morning light from the left, shallow depth of field.

Framing: 4:5 portrait, product in the lower third, clean negative space at the top for a headline.

Constraints: no text, no logos, no human faces, no brand marks on the product.

Campaign Copy and AI Product Visuals

STEP 5 OF 6

5

Iterate the image by changing ONE variable at a time - light, angle, styling or crop - never all at once.

PROMPT · paste into your AI assistant

Keep everything the same, change only the light: warm late-afternoon light from the right instead of morning light from the left.

Campaign Copy and AI Product Visuals

STEP 6 OF 6

6

Run the pre-publication check on the whole kit before anything goes live.

PROMPT · paste into your AI assistant

Check this campaign kit and list every issue as a numbered fix list:

1. Any claim not supported by products.csv.
 2. Any price, discount or date that is inconsistent between assets.
 3. Any line that breaks brand_voice.md.
 4. Anything that would need a disclosure because the image is AI-generated.
- Return only the issues and the fix for each - do not rewrite the campaign.

Campaign Copy and AI Product Visuals

Test it

Read the campaign aloud as a customer would meet it: subject line, then email, then a social caption, then the poster. The offer, the discount and the end date must be identical in all four, and every product claim must trace to products.csv. One inconsistency between assets means the kit is not ready to schedule.

Customer Data Privacy and Responsible AI

LAB 4

Before AI touches customer data, someone has to decide what it may see. You will classify the columns in the Bloom & Brew customer and transaction files, build an anonymised extract that is safe to paste into a public AI tool, draft a one-page AI usage policy for the store team, then red-team that policy to find what it fails to cover.

You'll build

An anonymised data extract that is safe to use with a public AI tool, plus a one-page AI usage policy for your store team.

Tools: ChatGPT or Claude, a spreadsheet, [labs/resources/customers.csv](#), [labs/resources/transactions.csv](#)

Customer Data Privacy and Responsible AI

STEP 1 OF 6



Open customers.csv and transactions.csv and mark every column that could identify a real person.

Customer Data Privacy and Responsible AI

STEP 2 OF 6

2

Ask the assistant to classify the columns - paste only the header row, never the customer data itself.

PROMPT · paste into your AI assistant

Role: you are a data protection adviser to a retail business in Singapore.

Context: these are the column headers of two customer files. I have deliberately not pasted any data.

customers.csv: customer_id, full_name, email, phone, postal_code, join_date, loyalty_tier, marketing_optin

transactions.csv: order_id, order_date, customer_id, sku, qty, unit_price, channel, store

Task: classify every column as SAFE, IDENTIFYING or SENSITIVE for use with a public AI tool.

Format: a table of column, classification, one-line reason, and the action to take before any AI use.

Constraints: be conservative - if a column could re-identify someone in combination with another, say so.

Customer Data Privacy and Responsible AI

STEP 3 OF 6



3

In the spreadsheet, build the anonymised extract: delete the identifying columns, renumber the customers and keep only the month.

Customer Data Privacy and Responsible AI

STEP 4 OF 6

4

Draft the store's one-page AI usage policy from your own classification work.

PROMPT · paste into your AI assistant

Role: you are writing an internal policy for a six-outlet specialty retailer.

Context: staff use public AI assistants for product copy, customer replies and sales analysis. The column classification above is our starting point.

Task: write a one-page AI usage policy the store team will actually follow.

Format: sections for (1) what may never be pasted into an AI tool, (2) what must be anonymised first, (3) which outputs need human approval before use, (4) who to ask when unsure. Bullet points, plain English, no legal jargon.

Constraints: maximum 400 words; every rule must be checkable by a shop-floor supervisor.

Customer Data Privacy and Responsible AI

STEP 5 OF 6

5

Red-team the policy: ask the assistant to find the realistic ways a busy team would breach it.

PROMPT · paste into your AI assistant

You are a sceptical store manager under time pressure.

Task: list the five most likely ways this policy gets broken in a real store on a busy Saturday, and the smallest change to the policy that would prevent each one.

Format: a table of scenario, why it happens, policy fix.

Constraints: realistic retail scenarios only - no hypothetical attackers.

Customer Data Privacy and Responsible AI

STEP 6 OF 6



6

Add the human-in-the-loop rule: name who approves AI output before it reaches a customer, a price or an order.

Customer Data Privacy and Responsible AI

Test it

Search your anonymised extract for '@', for a phone-number pattern and for any full name from the original file. All three searches must return zero hits - that is what makes the extract safe to paste into a public AI tool. If anything is still found, your column list in step 3 was incomplete; fix it and search again.

Recap — Introduction to AI for Retail

1

You can now: brief an AI assistant with the five-part prompt pattern and choose the right assistant for a retail task

2

You can now: produce on-brand product descriptions, feature bullets and SEO metadata from a product data file

3

You can now: generate a complete promotional campaign and an AI product visual, then run a pre-publication check

4

You can now: anonymise retail customer data before AI use and set the store's responsible-AI rules

TOPIC 02

Applying AI to Retail Operations and Customer Experience

AI-powered customer service · personalised recommendations and promotions · demand forecasting and inventory planning · pricing, merchandising and sales analytics · everyday AI workflows

KEY CONCEPTS

Key Concepts — Applying AI to Retail Operations and Customer Experience

1

A service assistant grounded in your own FAQ, returns policy and product data answers accurately. An ungrounded one invents policy you will have to honour.

2

Personalisation starts with segmentation: group customers by how recently, how often and how much they buy, then let AI write the offer for each segment.

3

Demand forecasting turns sales history into a buying decision. AI is fastest at spotting trend, seasonality and the slow movers quietly tying up cash.

4

Pricing and markdown calls need margin maths plus judgement. AI surfaces the candidates and the trade-offs; the merchant still makes the decision.

5

Sales analytics with AI is conversational: ask in plain English, then verify the number against the source data before you act on it.

6

A simple AI workflow is four parts - trigger, data, AI step, output - plus a human check. That is enough to automate a daily retail task with no code.

Hands-On Labs — Applying AI to Retail Operations and Customer Experience

Labs 5–6

- Build a Grounded Store Service Assistant
- Personalised Recommendations and Targeted Promotions

Labs 7–8

- Demand Forecasting and Inventory Planning with AI
- AI for Pricing, Merchandising and Sales Analytics

Lab 9

- Build a Simple AI Workflow for a Daily Retail Task

Build a Grounded Store Service Assistant

LAB 5

An assistant that has read your returns policy answers correctly; one that has not will invent a policy you then have to honour. You will build a Bloom & Brew service assistant grounded in the store policies, FAQ and product catalogue, give it tone and escalation rules, then test it against real customer enquiries - including one your policy does not cover.

You'll build

A store customer-service assistant grounded in your own policies and FAQ, with a tested escalation rule for anything it cannot answer.

Tools: ChatGPT Projects or Custom GPTs, or Claude Projects; labs/resources/store_policies.md, faq.md, products.csv, customer_feedback.csv

Build a Grounded Store Service Assistant

STEP 1 OF 6



Create a new Project (Claude) or Custom GPT (ChatGPT) named 'Bloom & Brew Service Assistant'.

Build a Grounded Store Service Assistant

STEP 2 OF 6



2

Add `store_policies.md`, `faq.md` and `products.csv` as the assistant's knowledge - this is what grounds it.

Build a Grounded Store Service Assistant

STEP 3 OF 6

3

Write the system instructions: role, tone, the grounding rule and the escalation rule.

PROMPT · paste into your AI assistant

You are the customer service assistant for Bloom & Brew, a specialty home, gift and coffee retailer with six outlets and an online store.

Answer ONLY from the attached store policies, FAQ and product catalogue.

If the answer is not in those documents, say: 'I'll pass this to a store colleague who can help' and state what information the colleague will need.

Never invent a policy, a price, a delivery date or a stock level.

Tone: warm, direct, British English, no more than 120 words per reply, no exclamation marks.

Always end a reply about a return, refund or exchange with the exact policy line it is based on.

Build a Grounded Store Service Assistant

STEP 4 OF 6

4

Test it on six real enquiries taken from customer_feedback.csv, one at a time.

PROMPT · paste into your AI assistant

A customer bought the ceramic pour-over set 26 days ago, has the receipt, and wants to exchange it for a different colour. What do I tell them?

Build a Grounded Store Service Assistant

STEP 5 OF 6

5

Test the escalation path with a question your policy deliberately does not cover.

PROMPT · paste into your AI assistant

A customer is asking whether you will price-match a competitor's website. What do I tell them?

Build a Grounded Store Service Assistant

STEP 6 OF 6



6

Tighten one weak answer by fixing the instructions, not by correcting the reply in the chat.

Build a Grounded Store Service Assistant

Test it

Ask the assistant a question your store policies genuinely do not cover - competitor price matching, or a return at 45 days. It must decline, use your escalation wording and state what the colleague will need. If it invents a policy instead, it is not ready to face a customer: strengthen the grounding rule and re-test.

Personalised Recommendations and Targeted Promotions

LAB 6

Personalisation starts with segmentation, not with a clever message. You will group the Bloom & Brew customers by how recently, how often and how much they buy, define four segments with a targeted offer and channel for each, then build next-best-product recommendations for three individual customers and check that the maths behind each offer still leaves a margin.

You'll build

Four customer segments with a targeted offer, channel and message for each, plus next-best-product recommendations for three named customers.

Tools: ChatGPT or Claude, a spreadsheet, [labs/resources/transactions.csv](#), [labs/resources/products.csv](#)

Personalised Recommendations and Targeted Promotions

STEP 1 OF 6



Use the anonymised extract from Lab 4, not the raw customer file - the same rule applies to your own store data.

Personalised Recommendations and Targeted Promotions

STEP 2 OF 6

2

Ask for a recency, frequency and value summary per customer, and a proposed set of four segments.

PROMPT · paste into your AI assistant

Role: you are a CRM analyst for Bloom & Brew.

Context: the attached file is an anonymised transaction extract. Columns: order_id, order_month, customer_ref, sku, qty, unit_price, channel, store.

Task: summarise each customer by recency (months since last order), frequency (orders in the period) and value (total spend), then propose four segments that cover every customer with no overlap.

Format: (1) a table of the four segments with their rule, customer count and share of total revenue; (2) the ten highest-value customers with their recency, frequency and value.

Constraints: state the exact rule for each segment so I can reproduce it in a spreadsheet; flag any customer the rules do not capture.

Personalised Recommendations and Targeted Promotions

STEP 3 OF 6



3

Sanity-check the segments in the spreadsheet before you build offers on top of them.

Personalised Recommendations and Targeted Promotions

STEP 4 OF 6

4

Design the offer, channel and message for each segment in one pass.

PROMPT · paste into your AI assistant

Task: for each of the four segments, design the next campaign.

Format: a table with columns - segment, objective, offer, channel, subject line or opening line (max 45 characters), and the one metric that tells us it worked.

Constraints: a different objective per segment - do not discount to customers who are already buying at full price; every offer must name a product category that exists in products.csv.

Personalised Recommendations and Targeted Promotions

STEP 5 OF 6

5

Build next-best-product recommendations for three individual customers from their own purchase history.

PROMPT · paste into your AI assistant

Task: for customer_ref <C0xx>, <C0yy> and <C0zz>, recommend the next TWO products each.

Format: a table of customer_ref, what they have bought, recommended SKU, and the one-sentence reason a store colleague could say out loud.

Constraints: recommend only SKUs in products.csv that the customer has not already bought; no recommendation may rely on a product attribute that is not in the file.

Personalised Recommendations and Targeted Promotions

STEP 6 OF 6



6

Check the margin: price the deepest offer against cost in products.csv before anyone approves it.

Personalised Recommendations and Targeted Promotions

Test it

Take one next-best-product recommendation and check it against that customer's own rows in the extract: the recommended SKU must be a product they have NOT bought, and the stated reason must match what they actually purchased. If the reason cites a purchase that is not in their history, the recommendation was invented - re-run it with the customer's rows pasted directly into the prompt.

Demand Forecasting and Inventory Planning with AI

LAB 7

A forecast is only useful when it ends in a buying decision. You will take two years of Bloom & Brew monthly sales, have AI identify trend and seasonality by category, forecast the next three months for the top SKUs, then convert that into reorder points with lead time and safety stock - and check one of them by hand before anyone raises a purchase order.

You'll build

A three-month demand forecast, reorder quantities with safety stock, and a slow-mover list ready for the buying meeting.

Tools: ChatGPT or Claude, a spreadsheet, labs/resources/sales_history_monthly.csv, labs/resources/products.csv

Demand Forecasting and Inventory Planning with AI

STEP 1 OF 6



1

Open sales_history_monthly.csv and check what you have: 24 months of units by SKU, category and month.

Demand Forecasting and Inventory Planning with AI

STEP 2 OF 6

2

Ask for the pattern first - trend, seasonality and anomalies by category - before asking for any forecast.

PROMPT · paste into your AI assistant

Role: you are a demand planner for Bloom & Brew.

Context: the attached sales_history_monthly.csv holds 24 months of unit sales by SKU, category and month.

Task: describe the demand pattern for each category - underlying trend, seasonal peaks and troughs, and any month that looks like an anomaly rather than a pattern.

Format: a table of category, trend direction, peak months, trough months, and a one-line note on anomalies.

Constraints: base every statement on the data in the file; state clearly where 24 months is too short to be confident.

Demand Forecasting and Inventory Planning with AI

STEP 3 OF 6

3

Forecast the next three months for the top ten SKUs by volume, with the method stated.

PROMPT · paste into your AI assistant

Task: forecast units for the next three months for the ten highest-volume SKUs.

Format: a table of SKU, category, last three months actual, forecast for each of the next three months, and the method used in one line.

Constraints: state the seasonal uplift you applied and why; give a low and high case as well as the base case; do not forecast SKUs with fewer than six months of history - list those separately.

Demand Forecasting and Inventory Planning with AI

STEP 4 OF 6

4

Convert the forecast into a reorder decision using lead time and safety stock.

PROMPT · paste into your AI assistant

Task: convert the base-case forecast into a reorder plan.

Context: supplier lead time is 6 weeks; we hold 2 weeks of safety stock; current stock on hand is in products.csv.

Format: a table of SKU, forecast monthly demand, lead-time demand, safety stock, reorder point, stock on hand, and order now yes/no with the quantity.

Constraints: show the arithmetic for each reorder point in a formula column so I can check it.

Demand Forecasting and Inventory Planning with AI

STEP 5 OF 6

5

Find the cash problem at the other end: the slow movers and the overstocks.

PROMPT · paste into your AI assistant

Task: list the SKUs where stock on hand exceeds six months of forecast demand, and the SKUs with no sales in the last three months.

Format: a table of SKU, category, stock on hand, months of cover, value at cost, and a recommended action - markdown, bundle, return to supplier or discontinue.

Constraints: use cost_price from products.csv for the value column; total the cash tied up at the bottom.

Demand Forecasting and Inventory Planning with AI

STEP 6 OF 6



6

Recompute one reorder point by hand in the spreadsheet before the plan goes anywhere near a supplier.

Demand Forecasting and Inventory Planning with AI

Test it

Recompute one reorder point by hand: $(\text{monthly forecast} / 4.33) \times 6 \text{ weeks lead time}$, plus 2 weeks of safety stock at the same weekly rate. The AI's number must match yours within rounding. If it does not, ask it to show the assumption it used for that row - the mismatch is always in the units, not the maths.

AI for Pricing, Merchandising and Sales Analytics

LAB 8

The margin question a retailer actually asks is which products to mark down, which to leave alone, and what it costs. You will build a margin picture by SKU and category, identify markdown and price-increase candidates, model what a 15% markdown does to profit, then produce a one-page KPI summary with three recommended actions - verifying the headline number yourself.

You'll build

A margin and markdown review with a modelled price scenario, and a one-page sales KPI summary with three recommended actions.

Tools: ChatGPT or Claude, a spreadsheet, [labs/resources/products.csv](#), [labs/resources/transactions.csv](#)

AI for Pricing, Merchandising and Sales Analytics

STEP 1 OF 6

1

Attach products.csv and the anonymised transaction extract, and ask for the margin picture by SKU and category.

PROMPT · paste into your AI assistant

Role: you are a category manager for Bloom & Brew.

Context: products.csv holds cost_price, retail_price and stock_on_hand; the transaction extract holds units sold by SKU.

Task: build the margin picture.

Format: (1) a table by category of revenue, gross margin percent and gross margin value, sorted by margin value; (2) the ten SKUs contributing the most margin and the ten contributing the least.

Constraints: gross margin percent = $(\text{retail_price} - \text{cost_price}) / \text{retail_price}$; show the formula you used for each aggregate; flag any SKU where cost exceeds retail.

AI for Pricing, Merchandising and Sales Analytics

STEP 2 OF 6

2

Ask for markdown candidates and price-increase candidates, each with the reason from the data.

PROMPT · paste into your AI assistant

Task: identify (a) five markdown candidates and (b) five price-increase candidates.

Format: a table of SKU, category, current margin percent, units sold, months of cover, recommendation and the one-line reason from the data.

Constraints: a markdown candidate must have both high cover and low velocity; a price-increase candidate must have strong velocity and below-category margin; do not recommend a price change on a SKU with fewer than 20 units sold.

AI for Pricing, Merchandising and Sales Analytics

STEP 3 OF 6

3

Model the money: what a 15% markdown does to margin, and how many extra units it must sell to break even.

PROMPT · paste into your AI assistant

Task: model a 15% markdown on the five markdown candidates.

Format: a table of SKU, current price, marked-down price, current margin percent, new margin percent, margin value lost per unit, and the extra units needed to hold total margin flat.

Constraints: state the break-even uplift as a percentage of current units; show the arithmetic; do not assume any demand elasticity you have not been given.

AI for Pricing, Merchandising and Sales Analytics

STEP 4 OF 6

4

Write the category review narrative a merchant would actually read.

PROMPT · paste into your AI assistant

Task: write the category review for the two largest categories by revenue.

Format: for each category - what is working, what is not, the single biggest opportunity, and the risk if we do nothing.
Maximum 120 words per category.

Constraints: every claim must cite a number from the analysis above; no generic retail advice.

AI for Pricing, Merchandising and Sales Analytics

STEP 5 OF 6

5

Produce the one-page KPI summary for management, ending in three actions with an owner and a date.

PROMPT · paste into your AI assistant

Task: write a one-page sales KPI summary for the monthly management meeting.

Format: (1) five headline KPIs with the number and the direction; (2) three recommended actions, each with the expected impact, a suggested owner role and a by-when; (3) one risk to flag.

Constraints: maximum 350 words; plain English; no recommendation without a number behind it.

AI for Pricing, Merchandising and Sales Analytics

STEP 6 OF 6



6

Verify the headline number yourself before the summary is circulated.

AI for Pricing, Merchandising and Sales Analytics

Test it

Verify the gross margin the AI reports for one SKU: $(\text{retail_price} - \text{cost_price}) / \text{retail_price}$ from products.csv, worked out in the spreadsheet. It must match to one decimal place. Then recompute the total margin value for the largest category the same way. If either differs, the aggregation is wrong and every table built on it needs re-running from the corrected figures.

Build a Simple AI Workflow for a Daily Retail Task

LAB 9

The last step is turning today's prompting into something that runs tomorrow without you. You will build a no-code daily workflow in a spreadsheet - trigger, data, AI step, output, human check - that produces the Bloom & Brew daily sales summary and drafts replies to overnight customer reviews, then document it as a one-page SOP and prove it repeats on a second day of data.

You'll build

A working no-code daily-summary workflow - trigger, data, AI step, output, human check - documented as a one-page SOP your team can run.

Tools: Google Sheets or Excel, ChatGPT or Claude, [labs/resources/transactions.csv](#), [labs/resources/customer_feedback.csv](#)

Build a Simple AI Workflow for a Daily Retail Task

STEP 1 OF 6



Define the workflow on paper first: trigger, data in, AI step, output, human check - five boxes, one line each.

Build a Simple AI Workflow for a Daily Retail Task

STEP 2 OF 6

A green circle with a white number 2 inside, indicating the second step of the process.

Set up the sheet: one tab for the day's transactions, one for overnight feedback, one for the output.

Build a Simple AI Workflow for a Daily Retail Task

STEP 3 OF 6

3

Write the AI step as a fixed, reusable prompt with the day's data as the only thing that changes.

PROMPT · paste into your AI assistant

Role: you are the duty manager writing the Bloom & Brew daily trading summary.

Context: below are yesterday's transaction rows and yesterday's customer feedback. Nothing else.

Task: write the daily summary.

Format: (1) five bullets - revenue, units, best category, worst category, one thing that stands out; (2) any stock or service issue that needs attention today; (3) a draft reply to each piece of feedback rated 3 stars or below, maximum 60 words each, in our house tone.

Constraints: use only the rows below; if a number cannot be calculated from them, write NOT AVAILABLE; never promise a refund or a discount in a draft reply - offer to have a colleague make contact.

Build a Simple AI Workflow for a Daily Retail Task

STEP 4 OF 6



4

Run the workflow end to end on one day of data and time how long it takes.

Build a Simple AI Workflow for a Daily Retail Task

STEP 5 OF 6



5

Add the human check that has to happen before anything is sent, and name who does it.

Build a Simple AI Workflow for a Daily Retail Task

STEP 6 OF 6

6

Document the workflow as a one-page SOP so someone else can run it tomorrow.

PROMPT · paste into your AI assistant

Task: turn the workflow I have just built into a one-page standard operating procedure.

Format: purpose, when it runs, who runs it, the numbered steps with the exact prompt to paste, what to check before sending, and what to do when the output looks wrong.

Constraints: written for a duty manager who has never used AI before; maximum one page; no jargon.

Build a Simple AI Workflow for a Daily Retail Task

Test it

Run the whole workflow again on a SECOND day of data without changing a word of the prompt. If the output comes back in the same format, with the numbers matching what the Data tab shows and no invented figures, the workflow is repeatable and ready for your team. If you had to adjust the prompt to make day two work, it is still a manual task - fix the prompt, not the day.

Recap — Applying AI to Retail Operations and Customer Experience

1

You can now: deploy an AI customer-service assistant grounded in your own policies, FAQ and product data

2

You can now: segment customers from transaction data and build targeted offers and next-best-product recommendations

3

You can now: turn sales history into a demand forecast, reorder quantities and a slow-mover list

4

You can now: analyse margin, model a markdown scenario and produce a KPI summary with recommended actions

5

You can now: build and document a repeatable no-code AI workflow with a human check for a daily retail task



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

AI Landscape

Describe the AI landscape from generative AI to AI agents, and identify high-value AI use cases across the retail value chain.

2

AI Assistants

Use AI assistants such as ChatGPT, Claude, Microsoft Copilot and Google Gemini to complete everyday retail tasks with well-structured prompts.

3

Content & Visuals

Generate on-brand product descriptions, marketing copy and product visuals with AI, and check them before they reach a customer.

4

Responsible AI

Apply customer data privacy, governance and responsible-AI practices when using AI tools on retail data.

5

Service & Personalisation

Deploy AI for customer service, personalised recommendations and targeted promotions in a retail setting.

6

Analytics & Workflows

Apply AI to demand forecasting, inventory planning, pricing, merchandising and sales analytics, and build simple AI workflows for daily retail tasks.

Continuing Your Journey

1

Pick the single lab that saves you the most time and run it on your own store data next week.

2

Build your prompt library: every prompt that worked today, saved with the output format it produced.

3

Agree an AI usage policy with your team before AI touches customer data or pricing.

4

Explore the follow-on courses in AI, data analytics and automation at www.tertiarycourses.com.sg.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Your course material stays available at <https://lms-tms.tertiaryinfotech.com/>.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

THANK YOU FOR ATTENDING

Thank You!

You have briefed, grounded, verified and shipped AI output across nine retail tasks today - keep applying the same pattern to your own store.